

- DFS traversal printed the complete family tree, branch by branch, as expected.
 - Full Tree Order: Great-Great-Grandfather → Great-Grandfather → Grandfather → Father → You → Sibling → Uncle → Grand-Uncle
 - Lineage path to 'You': Great-Great-Grandfather --> Great-Grandfather --> Grandfather --> Father --> You
- Time Complexity:
- DFS Traversal: $O(V + E)$ where V = people, E = parent-child relationships.
 - Path Finding: $O(V)$ in the worst case.

Compared to BFS (level-order), DFS is a natural fit for genealogy tracing as it mirrors how historians follow one lineage as deep as records allow before backtracking.

GitHub Link:

https://github.com/egotpat-gif/DSAIL-Case-Studies/tree/main/DSA_CO3_DFS-SEARCH_CASESTUDY

Student Signature	Faculty Signature